

Workshop 5: Database and Users' Privileges

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Contents

1 Be positive! (5 min.)	1
2 Create the school database (30 min.)	1
3 Create the users teacher and adminstudentcourse (20 min.)	2
4 Create the users teacher and adminstudentcourse (20 min.)	2
5 If you are done... ..	2

In this workshop, you will work in a team on a table with many other students around SQL. It is a collective discussion to find a solution to the given problem in the database "school" (Figure 1), step by step by exchange and comments. Please use the MariaDB database with Docker as explained in lecture 00 (at the end). Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue...

1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

2 Create the school database (30 min.)

From the figure of the school database (Figure 1), create the database and the tables in your MariaDB server. If you have any question, please ask me.

Please remember to apply the rules to extract the different tables from this Entity-Relationship Diagram (ERD).

Once created, populate the tables with 5 Students, 3 Courses and 10 Enrollments (each student can be enrolled in several courses, and each course can have several students). You can invent the data as you want, but please make sure to make realistic data and to have some enrollments for each student and for each course. You can also invent the names of the students and the courses as you want. If you have any question, please ask me.

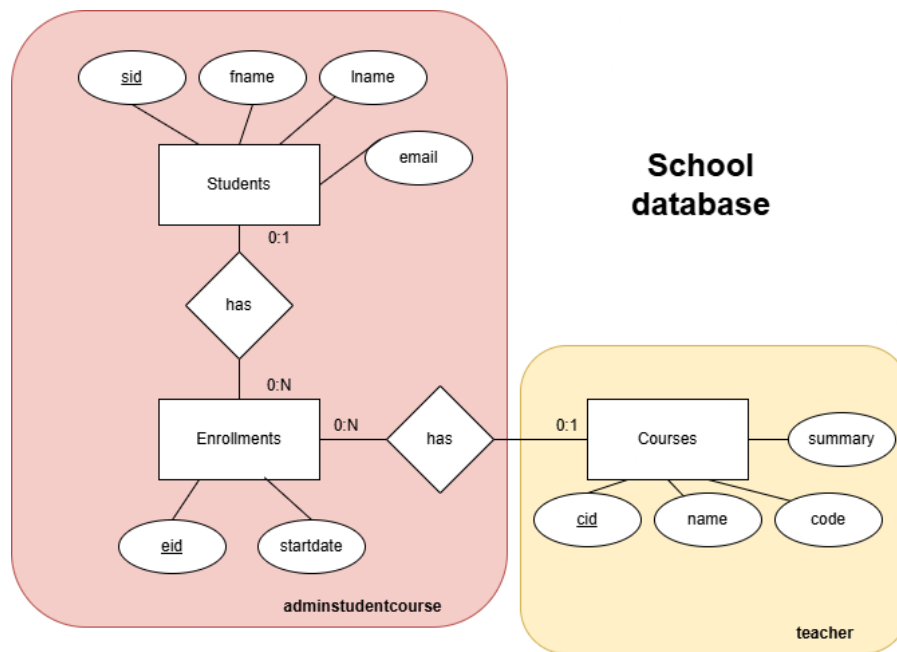


Figure 1: The school database.

3 Create the users teacher and adminstudentcourse (20 min.)

- The teacher can have access to the tables Students and Enrollments but ONLY for SELECT queries. But the teacher has ALL PRIVILEGES on the Courses table.
- The adminstudentcourse can have access to the Courses table ONLY for SELECT queries. But the adminstudentcourse has ALL PRIVILEGES on the Students and Enrollments tables.

4 Create the users teacher and adminstudentcourse (20 min.)

- Test your users by connecting to the database with their credentials and try to execute some queries on the tables. For example, try to SELECT from all the tables, try to INSERT, UPDATE or DELETE some data from the tables. You should see that the privileges you have set are working correctly. If you have any question, please ask me.
- Be sure that the teacher CAN change the course summary for instance while the adminstudentcourse CANNOT.
- Be sure that the teacher CANNOT change the name of a student for instance while the adminstudentcourse CAN.
- Be sure that the adminstudentcourse CANNOT change the name of a course for instance while the teacher CAN.
- Be sure that the adminstudentcourse CAN change the enrollments for the students in a COURSE instance while the teacher CANNOT.

5 If you are done...

In case you are done with the workshop very fast, you can work as a group on the next lab assignment. :) B+